

9 (**long**) A simple seesaw consists of a uniform rod of length $2l$ and mass m with a pivot at its center that allows it to rotate to form an angle θ with the horizontal. Two children, of masses m and $2m$, sit on the ends of the seesaw, with the heavier child touching the ground, and the seesaw making an initial angle θ_0 as shown in Fig. 7. The heavier child then applies an impulse P vertically upwards to launch the seesaw into motion.

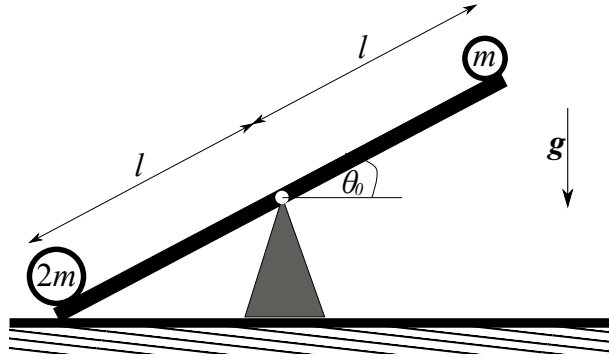


Fig. 7

- (a) Show that angular velocity, $\dot{\theta}_0$, of the seesaw directly after the impulse is: [5]

$$\dot{\theta}_0 = -\frac{3P \cos \theta_0}{10ml}.$$

- (b) Find the reaction impulse applied to the rod by the pivot, giving your answer in horizontal and vertical components. [5]

During the motion of the seesaw, sliding within the pivot joint causes a frictional torque τ of constant magnitude. When either child reaches the ground, they apply a vertical impulse that reverses their velocity.

- (c) Find the equation of motion for θ , and describe a forward Euler numerical integration scheme to predict the seesaw's subsequent motion over multiple oscillations. [8]

- (d) Sketch a graph of $\theta(t)$ for the case where the lighter child touches the ground twice. [6]

- (e) The initial impulse is chosen such that the lighter child just makes contact with the ground. Find an expression for the initial impulse. [6]